Unit - III

Maxwell's eq^ns in Integral & Differential forms

	Differential form	Integral form
1) Guass law in electricity	$\nabla \cdot E = \frac{\rho}{\epsilon_0}$	$\oint_S E \cdot ds = \int_V \frac{\rho}{\epsilon_0} dv$
2, Guass law for magnetism	$\nabla \cdot B = 0$	$\oint_S B \cdot ds = 0$
3) Faraday's law	$\nabla \times E = -\mu \frac{\partial H}{\partial t}$	$\oint_C E \cdot dl = -\int_S \frac{\partial B}{\partial t} \cdot ds$
4) Ampere's Law	$\nabla \times B = J + \epsilon \frac{\partial E}{\partial t}$	$\oint_C H \cdot dl = \oint_S \left(J + \epsilon_0 \frac{\partial E}{\partial t}\right) ds$ (or) $\oint_S H \cdot dl = \oint_S \left[J + \epsilon_0 \frac{\partial E}{\partial t}\right] ds$

physical significance of Maxwell eq^ns

→ I law shows that the total electric flux density 'D' through the surface enclosing a volume is equal to the charge density ρ with in the volume

→ II law tells us that the net mag. flux through a closed surface is zero — magnetic monopole does not exist

→ III law shows that electric field can also be generated by a time varying mag-field

→ IV equation states that magnetic motive force around a closed path is equal to the conduction current & displacement current.

↳ It means magnetic field is generated by a time varying electric field.

Derivation for Maxwell's eq^n's

Maxwells equations

I equation :

From Guass law of electricity

$$\oint E \cdot ds = \frac{q}{\epsilon_0} \quad \text{—} (1)$$

if ρ be the charge density & dv be the small volume

then charge density $= \rho = \frac{\text{charge}}{\text{volume}} = \frac{q}{dv}$

$$q = \rho . dv$$

for total charge $q = \int_V \rho \, dv$ —(2)

(2) in (1)

$$\oint E \cdot ds = \int_V \frac{\rho \, dv}{\epsilon_0} \quad \text{—} (3)$$

According to Guass divergence theorem

$$\oint A \cdot ds = \int_V (\nabla \cdot A) \, dv \quad \text{—} (4)$$

$$\int_V (\nabla \cdot E) \, dv = \int_V \frac{\rho}{\epsilon_0} \, dv$$

$$\boxed{\nabla \cdot E = \frac{\rho}{\epsilon_0}}$$

II eq^n: Guass law for magnetism

$$\oint B \cdot ds = 0$$

Transforming the surface to volume integral

$$\oint_V \nabla . B \, dv = 0 \qquad \boxed{\nabla \cdot B = 0}$$

III eq^n from Faraday's law

$$\oint_c E \cdot dl = \frac{-d\phi_B}{dt}$$

Magnetic flux density $B = \frac{\text{Flux}}{\text{area}(ds)}$

Flux $= B \cdot \text{Area}(ds)$

Total flux $= \oint_s B \cdot ds$

$$\int E \cdot dl = -\oint_s \frac{\partial B}{\partial t} ds \quad \text{— (5)}$$

Applying Stokes theorem

$$\oint E \cdot ds = \int_s (\nabla \times E) ds$$

$$\int_s (\nabla \times E) ds = -\oint_s \frac{\partial B}{\partial t} ds$$

$$\boxed{\nabla \times E = -\frac{\partial B}{\partial t}} \quad \text{(or)} \quad \text{curl } E = -\frac{\partial B}{\partial t}$$

IV eq^n from Ampere's law

$$\oint B \cdot dl = \mu_0 I \qquad \text{where } I = I_c + I_d$$

Let us consider current density $J = \frac{\text{current}}{\text{area}} = \frac{I}{ds}$

Total current $I = \oint_s J \cdot ds$

$$\oint B \cdot dl = \mu_0 \oint_s J \cdot ds$$

Applying Stokes theorem $\oint_s (\nabla \times B) ds = \mu_0 \oint_s J \cdot ds$

$$\boxed{\nabla \times B = \mu_0 J} \quad \text{— (6)}$$

But $J = \frac{\text{Total current}}{\text{area}} = \frac{i_c + i_d}{A} = \frac{i_c + A\epsilon_0 \frac{\partial E}{\partial t}}{A}$

$$J = \frac{i_c}{A} + \epsilon_0 \frac{\partial E}{\partial t} \quad \text{— (7)}$$

$j = \frac{i_c}{A}$ $(i_c = i_d = i)$

from (6) & (7)

$$\boxed{\nabla \times B = \mu_0 \left[j + \epsilon_0 \frac{\partial E}{\partial t} \right]}$$

Poynting vector – Energy density in EM waves

Let us consider an EM wave travelling along the x-axis with the magnetic & electric fields H & E confined in the transmission planes along z & y-axis respectively. This wave propagates along the direction of propagation vector $\vec{E} \times \vec{H}$. The vector $E \times H$ is known as poynting vector

~~Applying Divergent~~

Taking Divergence of poynting vector in free space

$\nabla \times E = -\frac{\partial B}{\partial t}$

$\nabla \times H = \mu\left(J + \epsilon\frac{\partial E}{\partial t}\right)$

Using this relation

$B = \mu_0 H$

$D = \epsilon_0 E$

i.e.,

$$\nabla \cdot (E \times H) = H \cdot (\nabla \times E) - E \cdot (\nabla \times H)$$

$$= -H \cdot \frac{\partial B}{\partial t} - E\left(\frac{\partial D}{\partial t}\right)$$

$$= -\left[\varepsilon_0 E \frac{\partial E}{\partial t} + \mu_0 H \cdot \frac{\partial H}{\partial t}\right]$$

$\frac{dx^2}{dt} = 2x\frac{dx}{dt}$

divide & multiply with '2'

$$= -\left[\frac{\varepsilon_0}{2} \cdot 2E\frac{\partial E}{\partial t} + \frac{\mu_0}{2} 2H \cdot \frac{\partial H}{\partial t}\right]$$

$$= -\left[\frac{\varepsilon_0}{2}\frac{\partial E^2}{\partial t} + \frac{\mu_0}{2}\frac{\partial H^2}{\partial t}\right]$$

$$\nabla \cdot (E \times H) = \frac{-\partial}{\partial t}\left[\frac{\varepsilon_0}{2}E^2 + \frac{\mu_0}{2}H^2\right]$$

Considering the surface 'S' bounds a volume V & integrating the above relation over the volume V, we get

$$\int_V \nabla\cdot(E\times H) = -\frac{\partial}{\partial t}\int_V\left[\frac{1}{2}\epsilon_0 E^2+\frac{1}{2}\mu_0 H^2\right]dv$$

applying divergence Theorem on the L.H.S of above eqn we get

$$\int_S (E\times H)\cdot ds = -\frac{\partial}{\partial t}\int_V\left[\frac{1}{2}\epsilon_0 E^2+\frac{1}{2}\mu_0 H^2\right]dv$$

The vector $P = E\times H$ is ~~interpreted~~ representing the amount of field energy passing through a ~~closed~~ unit surface area ~~of~~ in unit time normally to the direction of flow of energy. This is termed as poynting's Theorem. and vector P is called poynting vector.

$$\boxed{\oint_S (E\times H)\,ds = -\frac{\partial}{\partial t}\int_V\left[\underbrace{\frac{1}{2}\epsilon_0 E^2}_{\text{energy associated with electric field } u_E}+\underbrace{\frac{1}{2}\mu_0 H^2}_{\text{energy associated with magnetic field } u_m}\right]dv}$$

$u_E = \frac{1}{2}\epsilon_0 E^2$

$E^2 = \frac{2u_E}{\epsilon_0}$

$u_m = \frac{1}{2}\mu_0 H^2$

$B = \mu_0 H$

$H = \frac{B}{\mu_0}$

$u_m = \frac{1}{2}\frac{B^2}{\mu_0}$

as ~~B~~

$E/B = c$

& $c = \frac{1}{\sqrt{\mu_0\epsilon_0}}$

$u_m = \frac{1}{2\mu_0}\frac{E^2}{c^2}$

E^2